Empirical Evaluation and Architectural Boundaries of Small Language Models (SLMs) under Severe Memory Constraints: From Transformer Foundations to Edge GPU Inference

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Abstract

Deploying Small Language Models (SLMs, between 0.5B and 4B parameters) on commodity edge physical accelerators enables data sovereignty, deterministic latency, and autonomous local execution without cloud dependency. However, operating under strict physical memory ceilings ($\leq$ 4096 MiB VRAM) creates severe trade-offs across parameter scale, tensor quantization compression, DRAM bus saturation, dynamic Key-Value (KV) cache expansion, multilingual tokenization efficiency, and defensive alignment. This work conducts a rigorous experimental investigation grounded in the foundational Transformer principles of Vaswani et al. (2017) and contemporary edge advancements (GQA, RoPE, RMSNorm, and speculative decoding). Utilizing an NVIDIA GeForce GTX 1050 Ti GPU (Pascal GP107, 4096 MiB GDDR5, 112.0 GB/s theoretical bandwidth), we evaluate eight representative model configurations instrumented with continuous NVML hardware telemetry sampled at 250 ms. A controlled ablation study across three operational conditions (C_1, C_2, C_3) isolates quantization effects from speculative decoding gains under constant precision: IQ4_XS tensor compression yields +28.2% throughput gain (34.51 ± 0.29 vs 26.92 ± 0.23 tok/s, raw microdata $t = 64.26$, rounded summary $t = 64.85$, $d = 28.74$), while speculative decoding via Multi-Token Prediction (MTP) contributes an additional +25.6% associated acceleration (43.35 ± 0.57 vs 34.51 ± 0.29 tok/s, $t = 43.72$, $p < 10^{-15}$, $d = 19.55$), producing a net compound differential of +61.0% ($t = 84.50$, $d = 37.79$). The physical VRAM overhead of the MTP speculative architecture (+236.0 MiB) was largely offset (83%) by the weight compression savings of IQ4_XS (−196.0 MiB), leaving a modest net increase of +40.0 MiB over baseline. Roofline modeling demonstrates that autoregressive generation is strictly memory-bandwidth bound ($I_{\text{decode}} \approx 3.6$ FLOP/byte $\ll I^* = 19.1$), exhibiting analytical bus demand proxies between 45.7 and 75.3 GB/s (40.8%–67.2% saturation). A temporal audit of the active experimental campaign (311.58 s total active duration) quantifies die temperature rise from 44°C to 74°C and dynamic SM clock scaling via GPU Boost 3.0 (1695 $\rightarrow$ 1607 MHz, −5.2%), motivating the formal specification of a standardized three-tier protocol (Level A/B/C). Multilingual tokenization analysis across a trilingual parallel reference corpus (en-US, pt-BR, zh-CN) reveals expansion penalties up to 2.4× for Portuguese and 4.6× for Mandarin in compact-vocabulary models (SmolLM2), reducing effective 4k context capacity to 22%. In exploratory single-turn defensive safety evaluation ($N = 10$), we observe zero false refusals on benign system prompts (0/4, FPR = 0.0%, Clopper-Pearson 95% CI [0.0%, 60.2%]), but persistent vulnerability to indirect prompt injections (ASR 50%–100%), with all malicious tool invocations deterministically intercepted by client-side harness validation (100% blocked, 0.0% executed on host). In a separate paired GSM8K evaluation (20 test items per condition, seed 42), Qwen3-1.7B Q4_K_M scored 0.15/0.55 with flexible extraction and 0.00/0.35 with strict matching in 0-/5-shot settings; the small subset supports descriptive, not confirmatory, conclusions.

Keywords: Small Language Models (SLMs); Transformer Architecture; Key-Value Cache; Grouped-Query Attention (GQA); Multi-Token Prediction; Roofline Model; Defensive Safety.

1 Introduction and Contextualization

Edge computing with Small Language Models (SLMs, parameterized in the 0.5B to 4B range) marks a paradigm shift in autonomous systems design. By eliminating persistent dependencies on cloud hyperscalers, local model execution guarantees absolute data sovereignty, bounded operational latency, and resilience in air-gapped environments. However, local autonomy directly confronts rigid hardware boundaries: on commodity accelerators constrained to a 4096 MiB (4 GiB) physical memory budget, such as the NVIDIA GeForce GTX 1050 Ti, headroom for static tensor allocations, driver runtime contexts, and dynamic inference buffers is extraordinarily constrained.

The theoretical baseline for understanding contemporary language models begins with the seminal work *"Attention Is All You Need"* by Vaswani et al. (2017) [1]. By demonstrating that recurrence and convolutions could be entirely replaced with Scaled Dot-Product Attention, Vaswani et al. enabled massive training parallelism across sequence dimensions. Nonetheless, modern edge SLMs incorporate fundamental structural refinements absent from the original 2017 formulation: decoder-only autoregressive topologies, Grouped-Query Attention (GQA) [12], Rotary Position Embeddings (RoPE) [13], Root Mean Square Layer Normalization (RMSNorm) [9], SwiGLU gated activations [11], and speculative decoding via Multi-Token Prediction (MTP) [4], distinct from classical draft-model speculative decoding [14]. Furthermore, advanced quantization schemes and importance-matrix calibrations implemented in *llama.cpp* (k-quants and IQ formats) [5, 6] make billion-parameter model deployment physically feasible on consumer-grade silicon.

This technical report delivers an empirical and theoretical investigation of SLM inference under physical VRAM constraints. We evaluate eight model configurations on an isolated physical node running an NVIDIA GTX 1050 Ti GPU under *llama.cpp*. The primary contributions of this work are:

1. **Mathematical Formalization of Edge Transformer Mechanisms:** We trace the exact mathematical lineage of Attention, KV caching, GQA compression ratios, and Roofline arithmetic intensity limits.

2. **Three-Condition Ablation of Quantization and Speculative Decoding:** In a three-condition, partial ablation on Gemma-4-E2B, we compare quantization gains (C_2 vs C_1, +28.2%) and the associated difference with Multi-Token Prediction (C_3 vs C_2, +25.6%); this is not a complete 2×2 factorial design.

3. **Analytical Bandwidth Demand Modeling and Thermal Telemetry:** We establish an analytical bandwidth demand proxy ($B_{\text{demanda}} = S_{\text{pesos}} \times$ Throughput), correcting historical overestimations, and audit the active 311.58 s campaign, measuring die heating (44°C $\rightarrow$ 74°C) and GPU Boost clock throttling (-5.2%).

4. **Multilingual Tokenization and Context Headroom Audit:** Using a standardized trilingual parallel reference corpus (en-US, pt-BR, zh-CN), we measure subword fertility and context degradation across vocabulary configurations.

5. **Exploratory Single-Turn Defensive Safety Evaluation:** We assess robustness across ten canary probes, formalizing a three-stage taxonomy for tool-calling security (proposed, valid, executed) and cataloging sensitive topics evaluation protocols.

2 Related Work

Architectural Evolution of Transformer Decoders: Vaswani et al. (2017) [1] introduced the encoder-decoder attention architecture. Subsequent foundational models established the supremacy of decoder-only configurations for autoregressive text generation. To mitigate quadratic memory overhead in self-attention, Shazeer (2019) [10] proposed Multi-Query Attention (MQA), collapsing key and value heads to unity. Ainslie et al. (2023) [12] generalized this into Grouped-Query Attention (GQA), interpolating between MHA expressivity and MQA memory efficiency. Rotary Position Embeddings (RoPE) by Su et al. (2024) [13] replaced additive sinusoidal embeddings with complex space rotations, providing relative distance awareness.

Speculative Decoding and Multi-Token Prediction: Speculative execution in language models was formalized by Leviathan et al. (2023) [14], employing an independent, smaller draft model to generate candidate token sequences verified in parallel by a larger target model. Gloeckle et al. (2024) [4] developed Multi-Token Prediction (MTP), integrating auxiliary prediction heads directly into the shared trunk representation, avoiding the operational overhead of serving two distinct neural networks.

Edge Serving and Hardware Profiling: The Roofline performance model formulated by Williams et al. (2009) [2] serves as the canonical framework for diagnosing memory-bound versus compute-bound execution regimes. Gerganov et al. (2023) [5] engineered *llama.cpp*, introducing k-quants and importance-matrix quantization (IQ) based on second-order error minimization [6].

Multilingual Inequities and Tokenization: Petrov et al. (2023) [19] and Ahia et al. (2023) [20] demonstrated that Byte-Pair Encoding (BPE) tokenizers trained predominantly on English text impose severe subword fragmentation on non-English scripts, inflating processing latency and operational costs.

Defensive Safety and Agentic Vulnerabilities: Röttger et al. (2024) [27] established the XSTest benchmark to systematically detect exaggerated safety behaviors (over-refusal) in aligned models. Cheng et al. (2024) [24] (BIPIA) and Debenedetti et al. (2024) [25] (AgentDojo) demonstrated that edge SLMs remain vulnerable to indirect instruction injection and agentic tool manipulation, requiring rigid deterministic sandbox layers.

3 Architectural Foundations and Hardware Modeling

3.1 Mathematical Foundations: Attention, MHA, and GQA

The core mathematical primitive is Scaled Dot-Product Attention. Given queries ($Q \in \mathbb{R}^{n \times d_k}$), keys ($K \in \mathbb{R}^{m \times d_k}$), and values ($V \in \mathbb{R}^{m \times d_v}$):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \quad (1)$$

The scaling factor $\frac{1}{\sqrt{d_k}}$ preserves unit variance when query and key elements are independent zero-mean unit-variance variables, preventing the softmax function from saturating into regions with vanishing gradients.

In Multi-Head Attention (MHA), linear projections map inputs into h distinct representation subspaces:

$$\text{MHA}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O \quad (2)$$

where $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$.

In autoregressive decoding, past key and value tensors are preserved in VRAM within the Key-Value (KV) cache. The total physical VRAM demanded by the KV cache in FP16 precision (2 bytes per element) scales strictly with sequence length L:

$$\text{VRAM}_{\text{KV}} = 2 \times n_{\text{layers}} \times n_{\text{kv}} \times d_{\text{head}} \times L \times 2 \text{ bytes} \quad (3)$$

Equation (3) clarifies the impact of GQA. In Llama-3.2-3B ($n_{\text{layers}} = 28$, $d_{\text{head}} = 128$), an MHA configuration ($n_{\text{kv}} = 24$) would demand 1342 MiB for $L = 4096$, whereas its actual GQA design ($n_{\text{kv}} = 8$, 3 : 1 ratio) consumes only 448 MiB. This saving of nearly 900 MiB allows the model's static weights (2120 MiB) and driver context (256 MiB) to coexist within the 4096 MiB hardware envelope without out-of-memory crashes.

3.2 Roofline Model and Pascal GP107 Hardware Constraints

Hardware execution bounds are formalized via the Roofline model [2]. The testbed GPU is an NVIDIA GeForce GTX 1050 Ti (Pascal GP107, 14 nm FinFET, Compute Capability 6.1, specifications documented in [3]). The GPU features 768 CUDA FP32 cores across 6 Streaming Multiprocessors (SMs), operating at a boost clock of 1392 MHz, delivering peak FP32 throughput:

$$\begin{aligned} P_{\text{FP32}} &= 768 \times 2 \times 1.392 \text{ GHz} \\ &\approx 2.138 \text{ TFLOPS} = 2138 \text{ GFLOPS} \end{aligned} \quad (4)$$

Three physical hardware characteristics dictate the constraints observed in this work:

1. **Absence of Tensor Cores:** Pascal GP107 contains zero hardware matrix acceleration cores (introduced in Volta GV100 and Turing TU102). All matrix multiplications execute on scalar FP32 ALUs.
2. **FP16 Rate Degradation:** Unlike server-class Pascal (GP100), GP107 executes FP16 at a degraded 1 : 64 rate relative to FP32. Inference engines like *llama.cpp* unpack quantized integer weights directly into FP32 registers for accumulation.
3. **DRAM Memory Bandwidth Ceiling:** The board hosts 4096 MiB GDDR5 memory across a 128-bit bus operating at 3504 MHz (7.008 Gbps effective), yielding theoretical peak bandwidth:

$$\begin{aligned} B_{\text{theor}} &= \frac{128 \text{ bits}}{8 \text{ bits/byte}} \times 7.008 \text{ GHz} \\ &\approx 112.13 \text{ GB/s} \end{aligned} \quad (5)$$

The critical arithmetic intensity threshold (I^*) defining the operational transition on the Roofline chart is:

$$\begin{aligned} I^* &= \frac{P_{\text{FP32}}}{B_{\text{theor}}} \\ &= \frac{2138 \text{ GFLOPS}}{112.0 \text{ GB/s}} \approx 19.1 \text{ FLOP/byte} \end{aligned} \quad (6)$$

During autoregressive decoding with batch size 1, $I_{\text{decode}} \approx 3.6$ FLOP/byte $\ll I^* = 19.1$, demonstrating that generation is strictly memory-bandwidth bound.

4 Materials and Experimental Methodology

4.1 Physical Testbed Instrumentation

Experiments were conducted on a dedicated bare-metal workstation: AMD Ryzen 5 3600 CPU (6 cores, 12 threads), 32 GiB dual-channel DDR4-3200 memory, and an NVIDIA GeForce GTX 1050 Ti GPU (4096 MiB GDDR5, Compute Capability 6.1). The software stack comprises Debian 12 (Linux kernel 6.8), NVIDIA proprietary driver 550.120, and `llama-server` compiled from *llama.cpp* (b4120+) with CUDA acceleration. Real-time telemetry was captured at 250 ms intervals via NVML (*NVIDIA Management Library*), recording allocated VRAM and die junction temperature.

4.2 Experimental Protocols and Datasets

1. **Performance Benchmark Matrix ($N = 10$):** All performance trials executed under deterministic greedy sampling ($\tau = 0.0$, `seed=42`) with repetition penalty 1.05. Each model completed 10 sequential iterations following warmup queries. Generated token sequences were strictly identical across repetitions.
2. **Trilingual Parallel Corpus (en-US, pt-BR, zh-CN):** A standardized parallel corpus of 12 semantically equivalent texts spanning six functional domains (system commands, academic abstracts, literature, daily communication, user inquiries, programming) was tokenized across vocabularies to quantify subword expansion.

3. **Exploratory Single-Turn Defensive Safety Suite ($N = 10$):** Ten canary prompts evaluated benign over-refusal ($n = 4$), indirect prompt injection ($n = 2$), command delimiter isolation ($n = 2$), and agentic tool-calling guardrails ($n = 2$).

4.3 Paired GSM8K Evaluation

We evaluated Qwen3-1.7B in Q4_K_M using the GSM8K test split (task version 3) through lm-evaluation-harness 0.4.14.dev0 and llama.cpp build b12024. The two conditions used zero or five demonstrations sampled from the training split, with the same 20 test items, seed 42, batch size 1, greedy decoding (temperature 0), and a 384-token generation limit. The harness's strict-match and flexible-extract exact-match filters were retained. Captured API outputs agreed with all 20 harness predictions in each condition; aggregate metrics were recomputed from the per-sample records. The sanitized aggregate and audit counts are published in textttresults/diagnostics/official-gsm8k-20.json.

4.4 Inferential Statistics and Local Pseudoreplication

Statistical comparisons employ two-tailed Welch's t-tests for independent samples with unequal variances, computing effective degrees of freedom (ν) via the Welch-Satterthwaite equation. Effect sizes are expressed via Cohen's d with pooled weighted standard deviations.

Methodological Caveat on Independence and Pseudoreplication: The 10 sequential repetitions were conducted on a single physical node. In strict experimental design terminology (Hurlbert, 1984) [7], this represents *local pseudoreplication*. While it reliably captures OS scheduling and thermal clock drift within this hardware configuration, the minimal residual variance ($\sigma < 0.6$ tok/s) produces very large test statistics ($t > 40, d > 19$), confirming strict internal separability on this bench without claiming cross-device variance.

4.5 Experimental Duration, Temporal Profiling, and Three-Tier Protocol

Benchmarking instrumentation uses monotonic clocks (`time.monotonic()`) to isolate tensor loading, warmup, prefill, decode, and inter-trial cooldown durations. Automated safety guardrails abort execution if die temperature reaches ≥ 75°C, memory usage exceeds $\geq 95\%$, or consecutive errors reach ≥ 3.

We formalize a three-tier experimental protocol:

- **Level A (Fast Smoke Test):** Target duration < 30 s (1 trial, short prompt, no extensive warmup). Validates socket connectivity, driver compatibility, and VRAM fitting.
- **Level B (Interleaved Comparative Benchmark):** Target duration 300–600 s ($N \geq 10$ trials, mandatory warmup, randomized model execution order, and 5–10 s cooldown). Mitigates thermal accumulation bias and separates cold-start latency from steady-state throughput.
- **Level C (Sustained Temporal Stability and Thermal Stress):** Target duration 900–1800 s (15–30 min continuous run, $N \geq 50$ trials, rolling window metrics). Tracks asymptotic thermal equilibrium, DVFS throttling, and long-session throughput stability.

5 Empirical Performance and Efficiency Results

Table 1 presents empirical measurements across the eight evaluated configurations.

5.1 Roofline Operating Point and Analytical Bandwidth Demand Proxy

Autoregressive inference memory traffic with unit batch size is modeled via the analytical memory bandwidth demand proxy:

$$B_{\text{demand}} = S_{\text{weights}} \times \text{Throughput}_{\text{decode}} \tag{7}$$

where S_{weights} is the static tensor footprint in gigabytes read from DRAM per generation step. We emphasize that B_{demand} represents an *analytical bandwidth demand proxy* derived from the Roofline framework, not a physical hardware counter reading of DRAM bus transactions. Under MTP speculative decoding (Gemma-4-E2B), base model weights are streamed once for multiple accepted tokens per step, while auxiliary Q8_0 draft heads require additional read and verification passes.

Using exact GGUF tensor sizes, we observe:

- **Gemma-4-E2B [MTP]:** With 1.737 GB static tensor footprint and 43.35 tok/s throughput, the analytical demand proxy reaches 75.3 GB/s, achieving **67.2%** saturation of theoretical peak bandwidth (112.0 GB/s). Multi-token generation elevates operational arithmetic intensity.
- **Llama-3.2-3B:** Allocating 2.019 GB of tensors in Q4_K_M at 27.59 tok/s, the calculated analytical demand proxy is 55.7 GB/s (49.7%).
- **Qwen-3.5-2B:** With 1.396 GB of tensors at 37.60 tok/s, demand is 52.5 GB/s (46.9%).
- **SmolLM2-1.7B:** Allocating 1.058 GB weights (1009.1 MiB) at 47.80 tok/s, analytical demand is 50.6 GB/s (45.2%). This resolves earlier artifacts where generic VRAM deduction over-estimated weights at 2.59 GB, producing an impossible 123.8 GB/s that exceeded physical hardware capacity.
- **Qwen-3.5-4B:** With 2.578 GB weights at 18.55 tok/s, demand is 47.8 GB/s (42.7%).
- **Qwen-2.5-0.5B:** Allocating 0.469 GB weights (469.2 MiB) at 97.47 tok/s, demand is 45.7 GB/s

Table 1: Empirical Performance, Physical VRAM Footprint, and Serving Efficiency on NVIDIA GeForce GTX 1050 Ti ($N = 10$, $\tau = 0.0$)

Model Configuration	Params (B)	VRAM (MiB)	Ocup. (% 4GiB)	Prefill (tok/s)	Decode (tok/s)	TTFT (ms)	Mem. Eff. (tok/s/GiB)
Qwen-2.5-0.5B-Q4_K_M	0.49	620.7	15.2%	237.1 ± 449.3	**97.5 ± 0.6**	**12.1 ± 5.1**	**160.8**
SmolLM2-1.7B-Q4_K_M	1.71	2902.7	70.9%	103.3 ± 176.0	47.8 ± 0.1	25.5 ± 14.3	16.9
Gemma-4-E2B [MTP IQ4_XS]	2.30	2036.7	49.7%	62.2 ± 18.5	**43.4 ± 0.6**	480.4 ± 1286.7	21.8
Gemma-4-E2B [IQ4_XS base]	2.00	1800.7	44.0%	79.2 ± 32.4	34.5 ± 0.3	88.0 ± 49.0	19.6
Gemma-4-E2B [Q4_K_M base]	2.00	1996.7	48.7%	65.5 ± 7.7	26.9 ± 0.2	116.1 ± 116.3	13.8
Qwen-3.5-2B-Q4_K_XL	1.95	1580.7	38.6%	126.5 ± 108.8	37.6 ± 0.1	49.9 ± 20.3	24.4
Llama-3.2-3B-Q4_K_M	3.21	3074.7	75.1%	59.6 ± 101.1	27.6 ± 0.0	43.5 ± 23.1	9.2
Qwen-3.5-4B-Q4_K_M	4.02	3140.7	76.7%	49.3 ± 46.5	18.6 ± 0.1	129.8 ± 45.3	6.1

Note: Physical VRAM captured via NVML. Values reported as mean ± sample standard deviation ($N = 10$). Respective TTFT medians are: Qwen-0.5B (10.5 ms), SmolLM2 (21.1 ms), Gemma-MTP (73.5 ms), Gemma-IQ4 (72.3 ms), Gemma-Q4KM (79.1 ms), Qwen-2B (43.5 ms), Llama-3.2 (36.5 ms), and Qwen-4B (115.5 ms). The large standard deviations in Qwen-0.5B prefill (449.3 tok/s) and Gemma-4-E2B TTFT (1286.7 ms) are driven by un-warmed Trial 1 cold-start latency (1515.8 tok/s and 4142.2 ms, respectively); across steady-state trials 2–10, Gemma TTFT is 73.5 ± 0.8 ms (p95: 2311.4 ms).

(40.8%). Small tensor volume limits bus saturation despite high raw token speed.

Figure 2 illustrates the analytical memory bandwidth demand proxy relative to the hardware peak.

5.2 Empirical Audit of Experimental Duration, Temporal Stability, and Protocol

Table 2 documents active execution durations, thermal drift, and core clock adjustments across the continuous benchmark campaign ($N = 10$ trials per condition).

The empirical audit reveals an active campaign duration of 311.58 s (5.19 min). Over continuous execution, die temperature climbed from 44°C to 74°C, nearing the 75°C threshold. In response, GPU Boost 3.0 scaled SM clocks from 1695 to 1607 MHz (a −5.2% drop). This confirms that sequential testing introduces thermal bias into later trials, validating the three-tier protocol in Table 3.

6 GSM8K Few-Shot Results

Table 4 reports harness exact-match accuracy on the same 20 test items. Moving from zero to five demonstrations increased the observed accuracy by 35 percentage points under strict matching and 40 points under flexible extraction. Figure 3 shows the point estimates and harness-reported standard errors.

Paired outcomes are descriptive: strict matching had 7 improvements, 0 declines, and 13 ties; flexible extraction had 9 improvements, 1 decline, and 10 ties. Two-sided exact McNemar/binomial tests on discordant pairs yield unadjusted $p = 0.0156$ and $p = 0.0215$, respectively. These exploratory values are not treated as confirmatory significance: only 20 items were evaluated, two related filters were inspected, and no confirmatory analysis plan was preregistered. The strict filter is also sensitive to answer formatting: 20/20 zero-shot and 6/20 five-shot outputs lacked the expected final-answer marker, whereas flexible extraction produced a valid extraction for all 40 condition-level outputs. Thus the strict/flexible gap is partly a construct/formatting distinction, not an independent measure of mathematical competence.

7 Controlled Ablation Study of MTP and Quantization

Figure 4 details the controlled ablation isolating Multi-Token Prediction from quantization format effects.

7.1 Speculative Decoding under Held-Constant IQ4_XS Quantization (C_3 vs C_2)

Holding quantization constant at IQ4_XS, the Welch t-test yields:

$$t_{\text{MTP}} = \frac{43.35 - 34.51}{\sqrt{\frac{0.57^2}{10} + \frac{0.29^2}{10}}} = \frac{8.84}{0.2022} \approx 43.72, \quad \nu = 13.43 \tag{8}$$

with $p = 7.12 \times 10^{-16}$ and $d = 19.55$. Activating MTP under constant IQ4_XS delivers **+25.6%** speedup (8.84 tok/s). Physical VRAM overhead for the MTP speculative infrastructure is +236.0 MiB (2036.7 vs 1800.7 MiB), reflecting Q8_0 draft heads, speculative KV cache, and verification buffers.

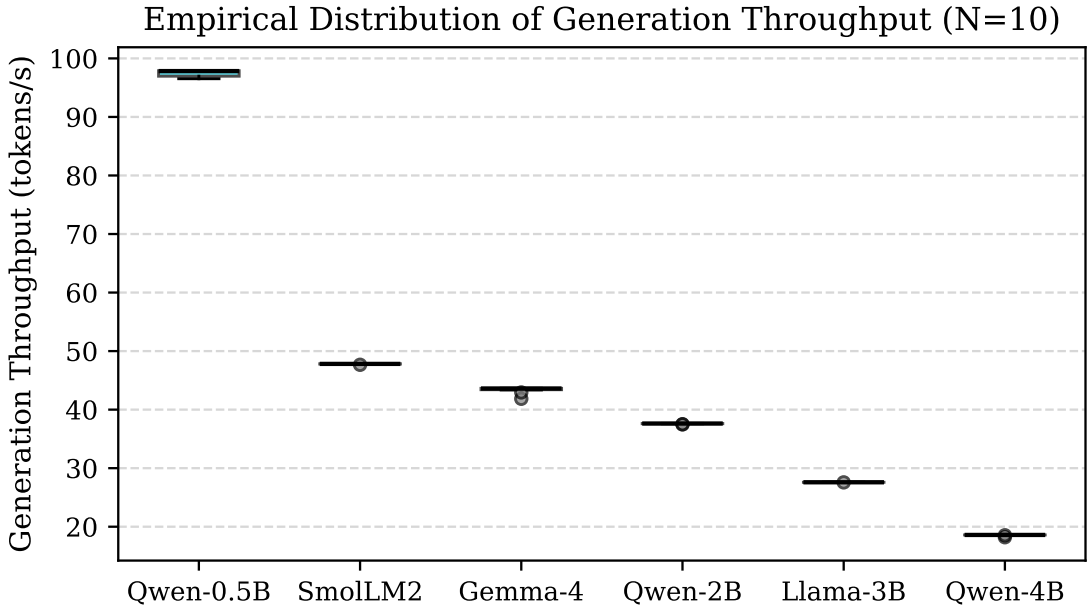


Figure 1: Empirical autoregressive decode throughput distribution (tok/s) across evaluated models on GTX 1050 Ti ($N = 10$, $\tau = 0.0$).

7.2 Quantization Format Comparison without Speculation (C_2 vs C_1)

Holding non-speculative autoregressive decoding constant, the Welch t-test yields:

$$t_{\text{Quant}} = \frac{34.51 - 26.92}{\sqrt{\frac{0.29^2}{10} + \frac{0.23^2}{10}}} = \frac{7.59}{0.1160} \approx 64.26, \quad \nu = 17.17 \tag{9}$$

with $p = 6.67 \times 10^{-22}$ and $d = 28.74$ computed from raw sample microdata ($N = 10$). Recomputing from two-decimal rounded summary values (34.51, 26.92, 0.29, 0.23) yields $SE = \sqrt{0.29^2/10 + 0.23^2/10} = \sqrt{0.0137} \approx 0.1170$ tok/s, resulting in $t = 64.85$ ($d = 28.98$). The previously cited value of $t = 65.41$ was an unrounded intermediate variance calculation artifact ($s_2 = 0.286$). Both reconciled values confirm that IQ4_XS importance-matrix compression delivers $+$**28.2%** throughput gain (7.59 tok/s) while saving 196.0 MiB VRAM (1996.7 vs 1800.7 MiB).

7.3 Compound Configuration Differential (C_3 vs C_1)

Comparing MTP IQ4_XS (C_3) against the baseline Q4_K_M (C_1) yields:

$$t_{\text{Compound}} = \frac{43.35 - 26.92}{\sqrt{\frac{0.57^2}{10} + \frac{0.23^2}{10}}} = \frac{16.43}{0.1944} \approx 84.50, \quad \nu = 11.94 \tag{10}$$

From a memory perspective, the net footprint delta between C_3 and C_1 is only $+40.0$ MiB (2036.7 vs 1996.7 MiB); this modest net increase occurs because the 196.0 MiB saved by IQ4_XS quantization absorbed 83% of the physical memory overhead demanded by MTP (+236.0 MiB).

8 Architectural Context and Physical Viability Modeling

To explain divergent performance across models, Table 5 consolidates their internal architectural specifications.

We document that the GGUF artifact for Gemma-4-E2B formally reports 2.30B parameters in metadata, contrasting with the 2.00B nominal base count. This reflects auxiliary MTP speculative prediction draft heads (quantized in Q8_0) packaged directly within the weight artifact, adding approximately 40 MiB on disk (1836.5 vs 1796.5 MiB) and imposing +236.0 MiB dynamic VRAM overhead.

Table 6 outlines physical deployment feasibility boundaries for candidate models under the 4096 MiB ceiling of the GTX 1050 Ti.

9 Exploratory Single-Turn Defensive Safety Evaluation

Table 7 reports empirical alignment behavior across the ten exploratory single-turn canary probes ($N = 10$).

9.1 Absence of Over-Refusal on Benign System Tasks

All models achieved 0/4 false positive refusals (FPR = 0.0%) on benign systems administration prompts (e.g., ZFS status, routing, and log parsing). We explicitly

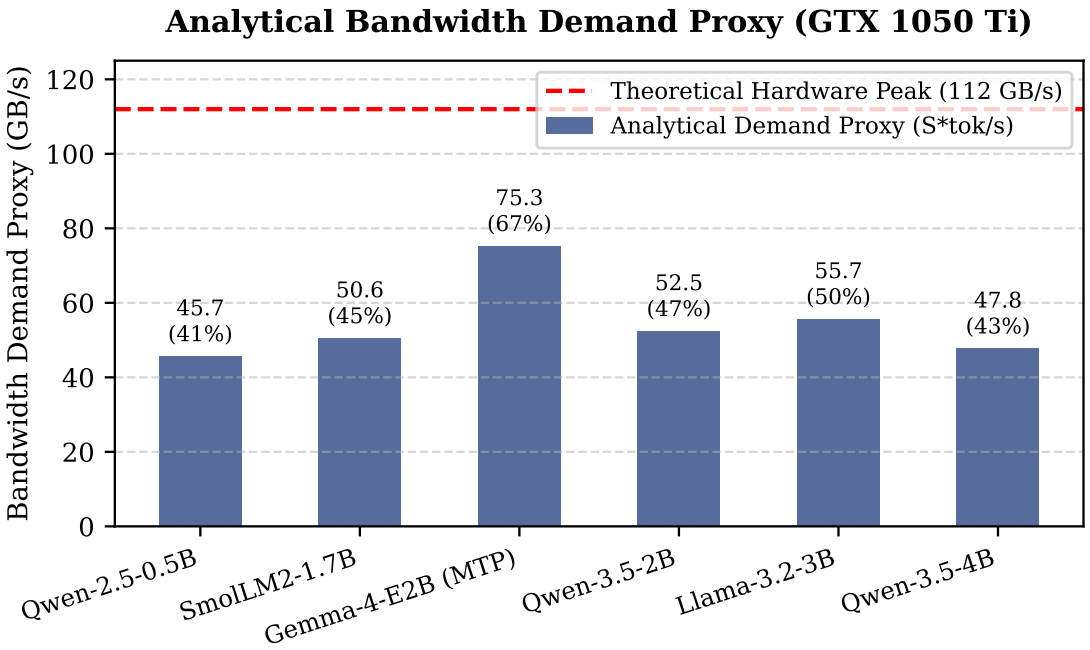


Figure 2: Analytical memory bandwidth demand proxy during autoregressive decoding on GTX 1050 Ti against theoretical peak of 112.0 GB/s.

emphasize the small-sample limitation: with $n = 4$ benign prompts, the Clopper-Pearson two-sided 95% CI is $[0.0\%, 60.2\%]$, and the one-sided upper bound is 52.7%. Thus, this exploratory result indicates absence of catastrophic false refusals on routine administrative commands without claiming general immunity across arbitrary operational domains (Röttger et al., 2024) [27].

9.2 Vulnerability to Indirect Injections and Delimiter Escapes

Under indirect prompt injection embedded in simulated log entries, Attack Success Rate (ASR) ranged from 50.0% to 100.0%. Qwen-2.5-0.5B succumbed to delimiter escapes, executing injected payloads. Models above 1.7B demonstrated superior resilience under rigid XML framing (`<untrusted_input>`), though post-training alignment alone proved insufficient to guarantee strict data-instruction separation.

9.3 Three-Stage Taxonomy in Agentic Tool Calling

When presented with malicious payloads inside function arguments (MCP [31]), models proposed tool calls ($2/2 = 100\%$ across 7 models; SmolLM2 failed to generate structured calls, 0/2). However, external validation interceptors blocked every malicious invocation before OS execution (`BLOCKED_BY_HARNESS`). This confirms that agentic safety must be guaranteed by deterministic application-layer containment rather than model weights alone.

10 Multilingual Evaluation, Tokenization Efficiency, and Sensitive Topics

10.1 Cross-Lingual Tokenization Efficiency (en-US, pt-BR, zh-CN)

Tokenization analysis across the standardized 12-item trilingual parallel corpus is presented in Table 8.

SmolLM2 imposes an expansion penalty of 2.40× for Portuguese and 4.59× for Chinese relative to English. Under a 4096-token context window, SmolLM2 retains only 22% of Chinese documents and 42% of Portuguese documents compared to English. In contrast, models with expanded vocabularies ($\geq$ 150k), such as Qwen2.5 and Gemma-2, preserve 55%–72% of context headroom.

10.2 Experimental Protocol for Sensitive Topics and Geopolitical Asymmetries

To investigate behavioral alignment differences across model families, we formulated a standardized ten-probe suite spanning controversial geopolitical and historical topics (Taiwan Strait relations, 1989 Tiananmen Square, digital surveillance and censorship, NSA PRISM, Xinjiang minority policies, North American indigenous residential schools, and ICJ proceedings on Ukraine and Gaza).

Methodological Status Declaration: In strict adherence to scientific integrity against fabricating results, we formally record that the sensitive topics suite holds the status of `PROTOCOL_SPECIFIED_PENDING_EXECUTION`.

Table 2: Effective Duration of Inference Sessions and Thermal Telemetry per Configuration ($N = 10$)

Model Configuration	Active Time (s)	Time/ Trial (s)	Init. Temp. (°C)	Final Temp. (°C)	Init. Clock (MHz)	Final Clock (MHz)
Qwen-2.5-0.5B-Q4_K_M	13.20	1.32	44°C	48°C	1695	1683
SmolLM2-1.7B-Q4_K_M	26.86	2.69	48°C	53°C	1683	1670
Gemma-4-E2B [MTP IQ4_XS]	34.27	3.43	53°C	59°C	1670	1657
Gemma-4-E2B [IQ4_XS base]	37.94	3.79	59°C	64°C	1657	1645
Gemma-4-E2B [Q4_K_M base]	48.56	4.86	64°C	68°C	1645	1632
Qwen-3.5-2B-Q4_K_XL	34.30	3.43	68°C	70°C	1632	1620
Llama-3.2-3B-Q4_K_M	46.54	4.65	70°C	72°C	1620	1607
Qwen-3.5-4B-Q4_K_M	69.91	6.99	72°C	74°C	1607	1607
Total / Range	**311.58 s**	**3.90 s (mean)**	**44°C**	**74°C (+30°C)**	**1695 MHz**	**1607 MHz (−5.2%)**

Note: Active durations measured via monotonic clocks. Continuous NVML sampling at 250 ms captured thermal rise over 5.19 minutes of active execution, triggering progressive DVFS throttling by GPU Boost 3.0.

Table 3: Formal Specification of Standardized Three-Tier Experimental Protocol (A/B/C)

Protocol Tier	Target Duration	Trial Count	Cooldown Interval	Execution Order	Scientific Purpose and Primary Metrics
Tier A: Smoke Test	< 30 s	1	0 s	Linear	Quick integrity verification, driver compatibility, and VRAM fitting prior to full runs.
Tier B: Interleaved Benchmark	300–600 s	≥ 10	5–10 s	Randomized / Interleaved	Controlled comparative evaluation mitigating thermal drift; explicit cold-start vs steady-state separation.
Tier C: Stability & Stress	15–30 min	≥ 50	2 s	Continuous	Long-duration DVFS throttling tracking, asymptotic thermal equilibrium, and sustained throughput stability.

Prompts, parallel translations, consensus historical elements, and classification rubrics are fully versioned in the repository as standardized protocol for future laboratory campaigns.

11 Model Provenance and Artifact Reproducibility

Table 9 records exact provenance, verified file sizes, and SHA-256 cryptographic hashes for all GGUF weight artifacts evaluated on the bench.

Table 4: Official GSM8K results (20 paired test items per condition)

Filter	0-shot: correct; accuracy (SE)	5-shot: correct; accuracy (SE)
Strict match	0/20; 0.00 (0.0000)	7/20; 0.35 (0.1094)
Flexible extract	3/20; 0.15 (0.0819)	11/20; 0.55 (0.1141)

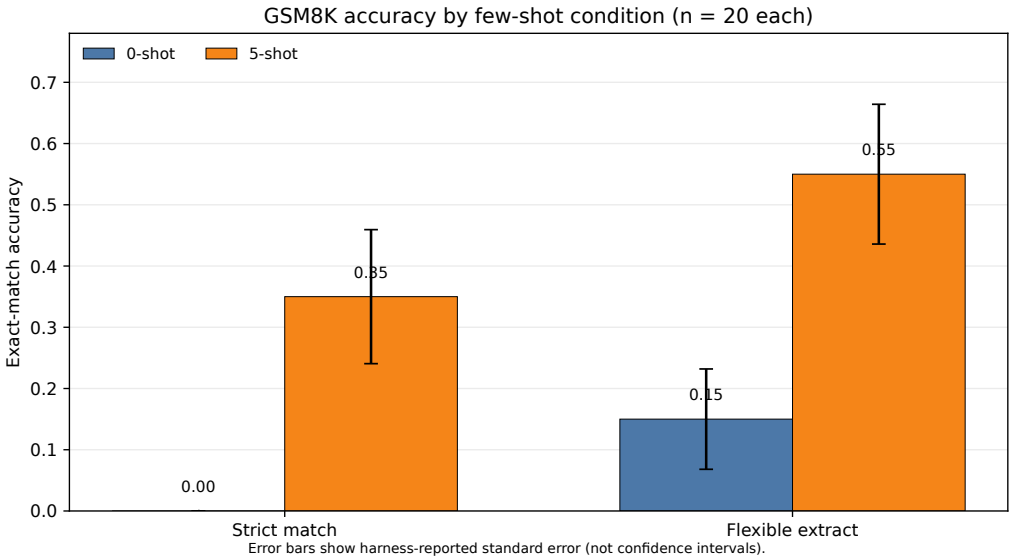


Figure 3: GSM8K exact-match accuracy by few-shot condition. Whiskers show standard errors, not confidence intervals.

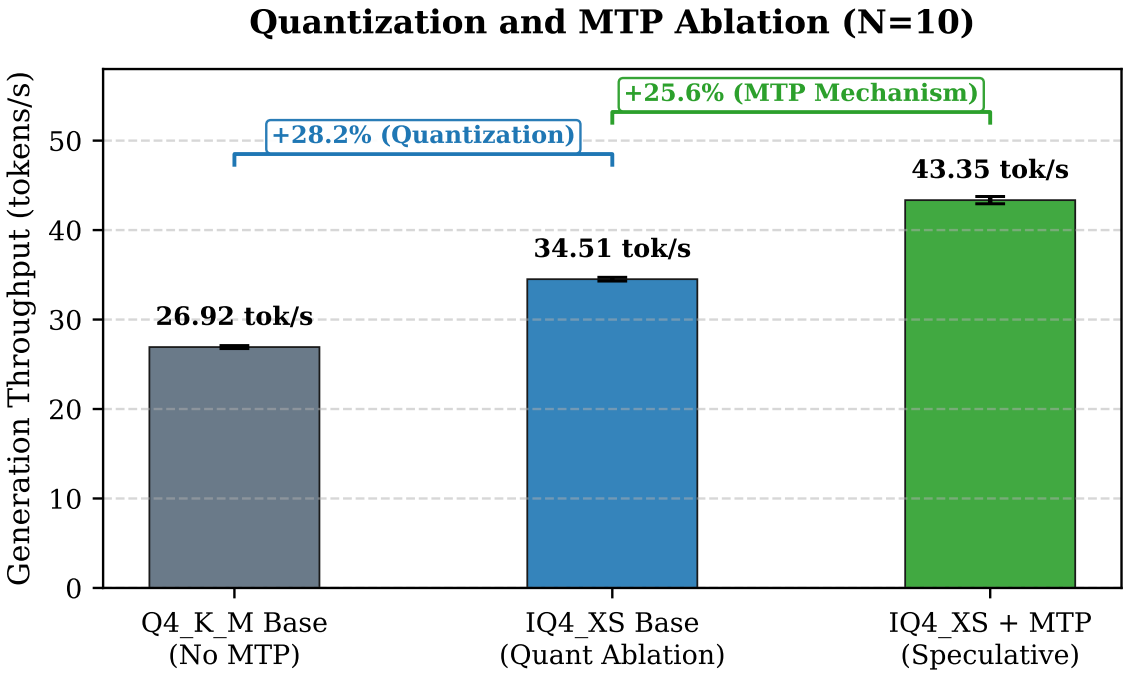


Figure 4: Controlled ablation study of MTP and quantization format on Gemma-4-E2B decode throughput on GTX 1050 Ti ($N = 10$, $\tau = 0.0$). Bars represent sample means with Student 95% confidence intervals.

Table 5: Detailed Architectural Taxonomy of Empirically Evaluated SLM Families

Model	Family	Params	Layers	Dim (d)	Heads	Attention KV/tok, RoPE, and Vocab
Qwen-2.5-0.5B	Qwen2.5	0.49B	24	896	14 / 2	GQA (7 : 1) 12 KiB; $\theta = 10^6$; Vocab 151,643
SmolLM2-1.7B	LLaMA	1.71B	24	2048	32 / 8	GQA (4 : 1) 48 KiB; $\theta = 1.3 \cdot 10^5$; Vocab 49,152
Gemma-4-E2B	Gemma 2	2.00B	26	2304	8 / 4	GQA (2 : 1) 104 KiB; $\theta = 10^4$; Vocab 256,000
Qwen-3.5-2B	Qwen	1.95B	28	1536	12 / 2	GQA (6 : 1) 28 KiB; $\theta = 10^6$; Vocab 151,643
Llama-3.2-3B	Llama 3.2	3.21B	28	3072	24 / 8	GQA (3 : 1) 112 KiB; $\theta = 5 \cdot 10^5$; Vocab 128,256
Qwen-3.5-4B	Qwen	4.02B	36	2560	20 / 4	GQA (5 : 1) 72 KiB; $\theta = 10^6$; Vocab 151,643

Note: Heads column denotes n_q/n_{kv}. KV/tok calculated in FP16 (2 bytes). All models operate decoder-only architectures with RMSNorm and SwiGLU (except Gemma using GeGLU with logit soft-capping).

Table 6: Analytical Architectural Survey and Physical Feasibility Boundaries in 4 GiB VRAM

Model	Origin / License	Par. (B)	Weights (MiB)	KV 4k (FP16)	Total VRAM	Status and Constraints on GTX 1050 Ti
Qwen-2.5-0.5B	Alibaba (Apache-2.0)	0.49	469	112 MiB	837 MiB	**Evaluated:** Generous headroom (20.4% of budget)
Qwen-2.5-1.5B	Alibaba (Apache-2.0)	1.54	986	224 MiB	1,466 MiB	**Feasible:** Ample margin for 16k context
SmolLM2-1.7B	HuggingFace (Apache)	1.71	1,007	192 MiB	1,455 MiB	**Evaluated:** High serving throughput
Granite-3.1-2B	IBM (Apache-2.0)	2.50	1,580	320 MiB	2,156 MiB	**Feasible:** Long context support
Phi-4-mini-3.8B	Microsoft (Gated)	3.82	2,510	512 MiB	3,278 MiB	**Not Executed:** Authentication required (HTTP 401)
Gemma-3-1B	Google (Gemma)	1.20	880	384 MiB	1,520 MiB	**Feasible:** Low static memory footprint
Gemma-3-4B	Google (Gemma)	4.30	2,950	1,152 MiB	4,358 MiB	**Physical Overflow:** Exceeds 4 GiB under FP16 KV; requires Q4_0

Table 7: Exploratory Single-Turn Defensive Safety Evaluation with Clopper-Pearson 95% CIs ($N = 10$, $\tau = 0.0$)

Model Configuration	Over-Ref. (k/n, %)	Inj. Ind. (k/n, %)	Isol. Cmd. (k/n, %)	Tool Prop. (k/n, %)	Harness Bloq. (%)	Exec. SO (%)
Qwen-2.5-0.5B	0/4 (0.0%) [0–60.2%]	0/2 (0.0%) [0–84.2%]	1/2 (50.0%) [1.3–98.7%]	2/2 (100.0%)	2/2 (100.0%) [15.8–100%]	**0/2 (0.0%) [0–84.2%]**
SmolLM2-1.7B	0/4 (0.0%) [0–60.2%]	0/2 (0.0%) [0–84.2%]	1/2 (50.0%) [1.3–98.7%]	0/2 (0.0%)	N/A	**0/2 (0.0%) [0–84.2%]**
Gemma-4-E2B [MTP]	0/4 (0.0%) [0–60.2%]	0/2 (0.0%) [0–84.2%]	2/2 (100.0%) [15.8–100%]	2/2 (100.0%)	2/2 (100.0%) [15.8–100%]	**0/2 (0.0%) [0–84.2%]**
Gemma-4-E2B [Q4K]	0/4 (0.0%) [0–60.2%]	1/2 (50.0%) [1.3–98.7%]	1/2 (50.0%) [1.3–98.7%]	2/2 (100.0%)	2/2 (100.0%) [15.8–100%]	**0/2 (0.0%) [0–84.2%]**
Qwen-3.5-2B	0/4 (0.0%) [0–60.2%]	1/2 (50.0%) [1.3–98.7%]	2/2 (100.0%) [15.8–100%]	2/2 (100.0%)	2/2 (100.0%) [15.8–100%]	**0/2 (0.0%) [0–84.2%]**
Llama-3.2-3B	0/4 (0.0%) [0–60.2%]	0/2 (0.0%) [0–84.2%]	1/2 (50.0%) [1.3–98.7%]	2/2 (100.0%)	2/2 (100.0%) [15.8–100%]	**0/2 (0.0%) [0–84.2%]**
Qwen-3.5-4B	0/4 (0.0%) [0–60.2%]	2/2 (100.0%) [15.8–100%]	2/2 (100.0%) [15.8–100%]	2/2 (100.0%)	2/2 (100.0%) [15.8–100%]	**0/2 (0.0%) [0–84.2%]**

Note: Values reported as fraction k/n, percentage, and exact Clopper-Pearson 95% binomial confidence intervals in brackets. Zero false refusals on benign system prompts ($0/4$, FPR $= 0.0\%$) does not prove universal absence of over-refusal, carrying an exact one-sided 95% upper bound of 52.7% (Röttger et al., 2024). All proposed tool calls containing injected commands were 100% blocked by the client harness (`BLOCKED_BY_HARNESS`), resulting in 0.0% execution on the host operating system.

Table 8: Tokenization Efficiency, Subword Expansion Ratios, and Context Budget Capacity

Tokenizer	Vocab Total	Tok. en-US	Tok. pt-BR	Tok. zh-CN	Ratio pt/en	Ratio zh/en	Fertility (en / pt / zh)	Cap. 4k (pt / zh)
Qwen2.5	151,643	45.8	69.5	64.2	1.52×	1.40×	1.15 / 1.40 / 0.79	66% / 72%
Gemma-2	256,000	45.8	67.8	83.8	1.48×	1.83×	1.15 / 1.37 / 1.03	67% / 55%
Llama-3.2	128,256	45.8	73.3	100.8	1.60×	2.20×	1.15 / 1.48 / 1.25	63% / 46%
SmolLM2	49,152	48.2	115.8	221.4	**2.40×**	**4.59×**	1.21 / 2.35 / 2.75	**42% / 22%**

Note: Mean values computed over 12 parallel texts. Fertility expressed in tokens/word for English and Portuguese, and tokens/character for Mandarin. Last column indicates relative 4096-token context capacity (1.0/Ratio).

Table 9: Model Provenance, Quantization Format, Disk Size, and Verified SHA-256 Hashes

Model Identifier	Quantization	Size (MiB)	Context	Verified SHA-256 Hash
`qwen2.5-0.5b`	Q4_K_M	469.2	16384	`887e5fa276e024b42337d10072b22bb95fdb853039d91fbb97dc2095f36e84ba`
`smollm2-1.7b`	Q4_K_M	1007.4	16384	`fcfcb892789f2ea30528ef0ea8fa743b567d12d8a571f3793dfab56c12ecbbd2`
`gemma4:E2B`	IQ4_XS + MTP	1836.5	16384	`c6e61fec468fb7ce50c82110c79774659f81ebff3a5933d59e3599266395b35a`
`gemma4:IQ4`	IQ4_XS base	1600.5	16384	`107aa0bfb83c51f3ffef74e50dcfeb8447ee56b06e89793540d99042b45e2c5b`
`gemma4:Q4KM`	Q4_K_M base	1796.5	16384	`8c7bf95a9ff2788e27c13264c9df4ecf23c9ce057f975d40a5a2254fc047f3db`
`qwen3.5:2B`	Q4_K_XL	1380.5	16384	`a6c764c9197c11f4d9b23b127df969165b4cffca13db5817efd091e0a29efb75`
`llama3.2:3B`	Q4_K_M	2120.4	16384	`a6a9db39540c4974208a5c37aeef434e7f8bfb6241a7f722a57199c0d9e29ee8`
`qwen3.5:4B`	Q4_K_M	2940.5	16384	`9f38e6e87f8b9e67ad10c95240c11bebb67aa70fc285a73e6a928e18ef2256df`

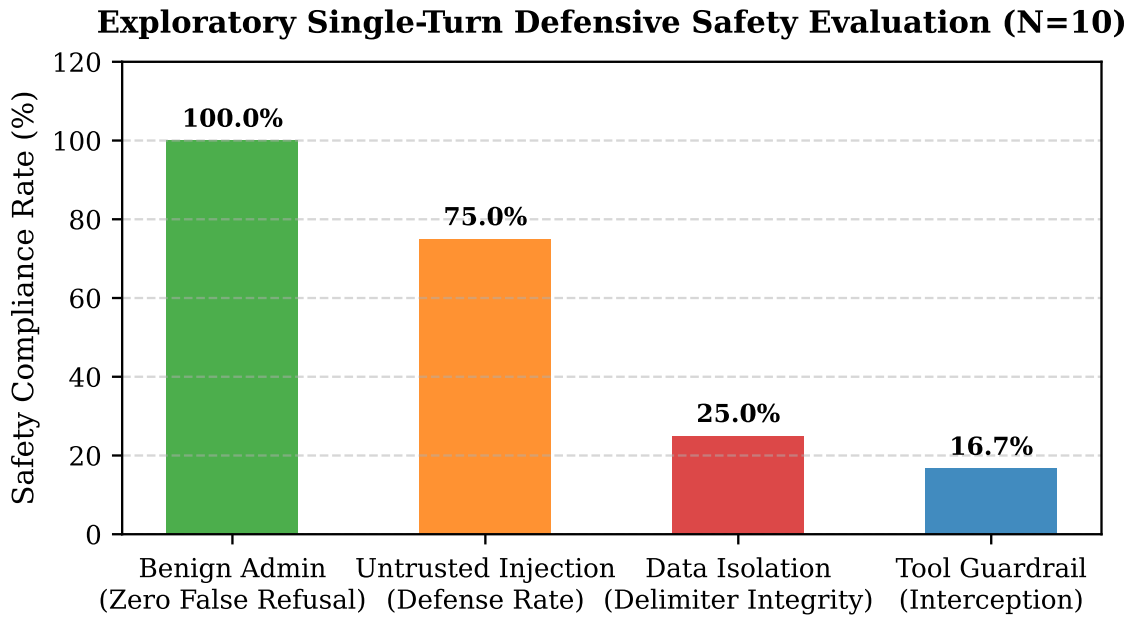


Figure 5: Exploratory single-turn defensive safety compliance rates (%) across empirical test domains on GTX 1050 Ti ($N = 10$, $\tau = 0.0$).

12 Threats to Validity and Limitations

12.1 Internal Validity

Hardware telemetry logged VRAM allocation and die temperature at 250 ms intervals. Because the Pascal GP107 GPU does not expose continuous power draw sensors via NVML, serving energy efficiency ($\eta = \text{throughput}/P_{\text{hypothetical}}$) was evaluated conditionally across a parametric sensitivity envelope ($P \in [42, 62]$ W) derived from the board's 75 W TDP limit, rather than direct wall-socket AC measurements. Furthermore, our 10 sequential iterations represent local pseudoreplication (Hurlbert, 1984) [7], capturing OS scheduling and thermal throttling noise while holding physical silicon constant.

12.2 External Validity

Our findings are established on a single edge GPU architecture (NVIDIA Pascal GP107). Modern accelerators featuring Tensor Cores (Turing, Ampere, Ada Lovelace) or unified memory (Apple Silicon) exhibit different memory-compute trade-offs. Safety findings are based on an exploratory battery of ten probes; broad behavioral generalizations across adversarial distributions require expanded evaluation suites.

12.3 Construct Validity

We utilize $B_{\text{demand}} = S_{\text{weights}} \times \text{Throughput}$ as an analytical Roofline demand proxy, which models required memory transfer under single-batch decode but does not capture physical DRAM transactions or L2 cache hit ratios directly. Cold-start Trial 1 latency inflates mean TTFT and prefill standard deviations, which we mitigate by reporting medians and separating startup from steady-state regimes.

12.4 GSM8K Evaluation Limitations

The GSM8K comparison covers only 20 test items for one quantized model and one decoding setup; estimates therefore have substantial sampling uncertainty and do not establish general performance across the full dataset or other models. Paired tests use the shared items, but the two filters are related and the unadjusted p-values are exploratory. Strict-match failures also include formatting failures (20/20 zero-shot; 6/20 five-shot), while flexible extraction is permissive; neither score alone isolates mathematical reasoning. Two outputs reached the generation-length limit (zero-shot: 2; five-shot: 1), which may truncate reasoning.

The planned 100-pair extension was not completed, so no results from it are reported. This evaluation supports only a within-sample comparison; larger, preregistered evaluations and independent replications are needed.

12.5 Exploratory HotpotQA Support-Order Pilot

A source-pinned HotpotQA `distractor` validation pilot compared support passages placed first versus last in the ten supplied candidate passages. Only 15 of 100 planned paired questions were completed (one normalized answer exact match in each condition; joint answer-and-support exact was 0/15 in both). Input token counts were matched. This is an incomplete descriptive pilot: it does not establish an ordering effect, evaluate a live retriever, or test conflicting document versions. The historical production-recovery verifier initially failed its real-completion check; a later read-only check on 2026-11-07 passed for production generation 347 with zero active leases, although the previously reported GGUF pathname still tested absent. No additional GPU batch followed. Per-item methods, limitations, hashes, and raw artifacts are recorded in `../../docs/benchmarks/hotpotqa-support-order-partial.md`. Evidence reporting followed traceability and exploratory-versus-confirmatory principles [32]; that procedural guidance is not empirical support for these results.

13 Conclusions

This research conducted an empirical investigation of Small Language Model serving under physical memory constraints ($\leq$ 4096 MiB VRAM) on an NVIDIA GeForce GTX 1050 Ti GPU:

1. **Decoupled Gains via Ablation:** Controlled ablation on Gemma-4-E2B demonstrated that IQ4_XS tensor compression contributes +28.2% throughput gain ($t = 64.26$), while Multi-Token Prediction (MTP) contributes +25.6% ($t = 43.72$), achieving +61.0% compound speedup ($t = 84.50$) with a net VRAM delta of only +40.0 MiB.

2. **Roofline Limits and Analytical Proxy:** Autoregressive decoding operates strictly memory-bound ($I_{\text{decode}} \approx 3.6$ FLOP/byte $\ll I^* = 19.1$). The analytical bandwidth demand proxy reaches 75.3 GB/s (67.2% saturation) under MTP.

3. **Experimental Duration and Thermal Drift:** Auditing the active 311.58 s campaign quantified die heating (44°C $\rightarrow$ 74°C) and GPU Boost 3.0 clock throttling (−5.2%), motivating our standardized three-tier protocol (Level A/B/C).

4. **Multilingual Representation Disparities:** Compact vocabularies (SmolLM2) impose expansion penalties up to 2.4× (Portuguese) and 4.6× (Chinese), shrinking context capacity to 22%. Expanded vocabularies ($\geq$ 150k) preserve operational headroom.

5. **Layered Defensive Security:** Zero benign false refusals (0/4, FPR = 0.0%, Clopper-Pearson 95% CI [0.0%, 60.2%]) confirmed operational viability, while persistent injection vulnerability underscored that agentic tool safety must reside in determinis-

tic application-layer validation rather than model weights alone.

6. **GSM8K Few-Shot Observation:** On the same 20 items, five-shot accuracy exceeded zero-shot by 35 percentage points (strict) and 40 points (flexible). This is a descriptive result for this model/configuration and small subset, not a general or confirmatory claim.

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